

**The Relationship Between Countermovement Jump Force Time Curve Waveforms and
Associated Kinetic Variables in Female Collegiate Basketball Division 1 Athletes**

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Background:

Countermovement jump (CMJ) testing is widely used to assess athletic performance qualities and monitor neuromuscular readiness. The morphology of the CMJ force-time curve—typically characterized as unimodal or bimodal—has gained attention as a potential indicator of underlying movement strategies. While previous research has investigated curve shapes in general populations, limited evidence exists in female collegiate basketball athletes.

Purpose:

To examine whether the CMJ force-time curve shape (unimodal vs. bimodal) is associated with differences in key CMJ performance metrics among Division I female basketball players.

Study Design:

Cross-sectional comparative study.

Methods:

Nine Division I female basketball athletes (five unimodal, four bimodal) were categorized based on consistent force-time waveform shape across multiple CMJ testing sessions using Hawk Dynamics force plates. Key kinetic variables—such as jump height, propulsive and braking force/power, time to takeoff (TTT), RSI-mod, and countermovement depth (CMD)—were compared between groups using independent t-tests.

Results:

Bimodal athletes exhibited significantly higher jump height ($p = 0.000$), relative braking force ($p = 0.018$), braking power ($p = 0.000$), propulsive power ($p = 0.014$), and RSI-mod ($p = 0.009$). They also showed longer braking, propulsive, and total movement durations (all $p = 0.000$) and

greater CMD ($p = 0.000$). In contrast, unimodal athletes demonstrated significantly greater average relative propulsive force ($p = 0.000$) and shallower countermovement depths.

Discussion

Results suggest distinct neuromuscular strategies between curve types. Bimodal athletes tend to rely on deeper, more temporally extended movement patterns, while unimodal athletes display stiffer, quicker strategies. Findings align partially with Peng et al. (2019), who reported similar performance advantages in bimodal jumpers, though some differences (e.g., TTT) may reflect sport-specific adaptations. Curve shape alone should not be interpreted as biomechanically superior; instead, it may serve as a context-specific profiling tool.

Practical Relevance:

The shape of the CMJ force-time curve provides insight into athlete-specific jumping strategies and can inform individualized programming. Both unimodal and bimodal profiles appear to support viable performance outputs, reinforcing the need for athlete-centered interpretations.

Key Terms:

Countermovement Jump, Force-Time Curve, Braking Phase, Propulsive Force, RSI-mod, Female Athletes, Basketball, Performance Profiling

What is known about this subject:

Previous studies have identified distinct unimodal and bimodal force-time curve profiles and associated them with different mechanical outputs, primarily in male and non-athlete populations.

What this study adds to existing knowledge:

This study provides evidence that the CMJ curve shape is associated with meaningful performance differences in female collegiate basketball athletes. It supports the use of waveform analysis as a viable tool for profiling and programming while cautioning against assuming superiority of any one curve type.

Introduction

The Countermovement Jump (CMJ) serves as a commonly used assessment tool employed by sports teams to evaluate critical performance metrics and monitor neuromuscular fatigue in athletes. Particularly within the context of basketball, the CMJ is favored for its time efficiency and its applicability to sport-specific movements. Typically executed with the hands placed on the hips or accompanied by an arm swing, the fundamental objective of the CMJ remains unchanged: to give performance specialists feedback regarding the athlete's performance.

In recent years, the morphology of the CMJ force-time curve has become a matter of extensive academic inquiry. Practitioners have observed divergent curve shapes among specific athletes. For instance, certain individuals exhibit double-peaked graphs, classified as bimodal, while others demonstrate graphs with a singular peak, referred to as unimodal curves. The underlying reasons for these variations in force-time curves have incited debate among both practitioners and researchers. One prevailing theory suggests that the recruitment patterns of specific muscles and tissues differ between individuals characterized as "unimodal jumpers" and those identified as "bimodal jumpers."

A research study investigating muscle-specific contributions to ground reaction force (GRF) profiles in collegiate basketball players revealed a significant correlation between the

activation of the soleus muscle and unimodal vertical ground reaction force (vGRF) profiles, whereas considerable activation of both the soleus and gastrocnemius muscles was associated with bimodal classifications (1). These distinctions suggest that athletes exhibiting unimodal jump curves may primarily engage in dorsiflexion, in contrast to bimodal athletes, who may engage in both dorsiflexion and plantar flexion.

Given these variations in force-time curve shapes, a pertinent inquiry arises for practitioners: Which jump profile is superior? This question has been the focus of scholarly exploration. In a study aimed at determining the biomechanical efficiency of vGRF profiles during the countermovement jump, researchers concluded that, irrespective of whether an individual exhibited a unimodal or bimodal profile, the most efficient jumping mechanics were linked to the peak force occurring at the lowest position of the countermovement (3). Efficient jumps were characterized by enhanced initial unweighting, expeditious braking phases, and elevated values across all five force metrics during the braking phase (Unweighting, Braking Force, Braking Power, Braking Rate of Force Development, and Force at the lowest position), ultimately resulting in superior jump heights and Reactive Strength Index (RSI) scores despite no statistically significant differences observed in peak propulsive power, average propulsive force, or duration of propulsion.

Despite the insights found from this research, a significant gap in understanding the interaction between curve shape and jump performance persists among female collegiate basketball athletes. A pertinent question remains: Does this research hold relevance for female basketball players, or do disparate patterns emerge? Accordingly, the present study aims to investigate the relationship between CMJ force-time curve shapes and key performance indicators, focusing specifically on CMJ metrics associated with theoretical athletic performance

in female collegiate Division 1 athletes, such as various deceleration (braking), propulsive, and neuromuscular efficiency metrics. The purpose of this study was to examine whether the countermovement jump (CMJ) force-time curve shape (unimodal vs. bimodal) is associated with differences in key CMJ performance metrics in female collegiate basketball athletes. We hypothesized that jump performance would differ between individuals with bimodal versus unimodal force-time curve profiles.

Methods

A cross-sectional study was conducted for this research, focusing on a sample of 9 Division I female collegiate basketball athletes. The inclusion criteria required participants to be healthy athletes actively engaging in practice and strength training during both the preseason and in-season periods. The study protocol involved conducting two to three jump sessions per week, with each session comprising two to three jumps. Averages were calculated for each session to assess performance metrics accurately. Jump testing was conducted over a five-week preseason training consisting of 20-hour weeks of NCAA-regulated activity. Each session included two to three CMJs per athlete. All jumps were performed using a Hawkin Dynamics dual force plate system sampling at 1000 Hz. Athletes completed jumps with hands on hips to minimize upper body variability and were instructed to jump “High and Fast” following a standardized quiet period. Force-time data were collected using the Hawkin Dynamics iPad app and later exported for analysis.

Jump waveform classification was determined via visual inspection of the vertical ground reaction force-time curve. Jumps were categorized as either unimodal (a single, clear peak during the propulsive phase) or bimodal (two distinct peaks during braking and propulsion, with the first peak happening at the transition point from braking to propulsion).

To assign athletes to groups, the most consistently displayed waveform across all testing sessions was used as the classification criterion. If an athlete demonstrated the same waveform in most sessions, they were placed into the corresponding group. This method allowed for individualized grouping based on repeated jump strategy rather than single-trial variability.

Key kinetic variables analyzed included: Jump Height (in), Average Relative Propulsive Force, Average Relative Braking Force, Peak Relative Braking Power, Peak Relative Propulsive Power, Time to Takeoff (s), Modified Reactive Strength Index (RSI-mod), and Countermovement Depth (in). Phase metrics were also analyzed, including the braking phase and the Propulsive phase. Descriptive statistics (mean $\pm$ standard deviation) were calculated for all variables.

The Shapiro-Wilk test was used to test for normality. All performance variables met the assumptions for a normal distribution ($p > 0.05$), allowing for the use of parametric tests. Independent t-tests were used to compare differences between unimodal and bimodal groups.

Results

A total of nine athletes were included in the analysis, with five classified as unimodal and four as bimodal jumpers based on consistent visual waveform inspection across sessions. Table 1 presents the descriptive statistics (mean $\pm$ SD) and between-group comparisons for key CMJ performance metrics. All data were normally distributed ($p > 0.05$). Significant differences ($p < 0.05$) were observed across several variables.

Athletes exhibiting bimodal force-time curves demonstrated significantly higher jump height (14.26 ± 1.5 in) than their unimodal counterparts (12.17 ± 1.3 in; $p = 0.000$). Bimodal athletes also produced substantially higher relative peak propulsive power (51.85 ± 4.8 W/kg vs.

49.75 $\pm$ 4.2 W/kg; $p = 0.014$) and displayed greater relative braking power (-24.9 ± 3.5 W/kg vs. -19.16 ± 3.7 W/kg; $p = 0.000$). Notably, bimodal athletes exhibited longer propulsive phases (0.256 ± 0.02 s vs. 0.221 ± 0.02 s; $p = 0.000$) and time to takeoff (0.703 ± 0.06 s vs. 0.641 ± 0.06 s; $p = 0.000$). They also produced significantly higher RSI-mod values (0.519 ± 0.06 vs. 0.488 ± 0.06 ; $p = 0.009$), and greater relative braking force (20.45 ± 1.6 N/kg vs. 19.75 ± 2.9 N/kg; $p = 0.018$).

Unimodal jumpers showed higher average relative propulsive force (21.07 ± 1.26 N/kg vs. 20.29 ± 0.97 N/kg; $p = 0.000$). Additionally, unimodal athletes demonstrated significantly shallower countermovement depths (CMD) than bimodal athletes ($p = 0.000$), highlighting a consistent pattern of reduced displacement during the preparatory phase of the jump.

Table 1. Descriptive Statistics for CMJ Performance Metrics by Force-Time Curve Classification

Metric	Unimodal (Mean $\pm$ SD)	Bimodal (Mean $\pm$ SD)	p-value
Jump Height (in)	12.17 $\pm$ 1.3	14.26.3 $\pm$ 1.5	0.000*
Relative Peak Propulsive Power (W/kg)	49.75 $\pm$ 4.2	51.85 $\pm$ 4.8	0.014*
Avg Relative Propulsive Force (N/Kg)	21.07 $\pm$ 1.26	20.29 $\pm$ 0.97	0.000*
Propulsive Phase (s)	0.221 $\pm$ 0.02	0.256 $\pm$ 0.02	0.000*
Relative Peak Braking Power (W/kg)	-19.16 $\pm$ 3.7	-24.9 $\pm$ 3.5	0.000*
Avg Relative Braking Force (N/kg)	19.75 $\pm$ 2.9	20.45 $\pm$ 1.6	0.018*
Braking Phase (s)	0.123 $\pm$ 0.02	0.141 $\pm$ 0.02	0.000*
Time to Takeoff (s)	0.641 $\pm$ 0.06	0.703 $\pm$ 0.06	0.000*

RSI-mod	0.488 ± 0.06	0.519 ± 0.06	0.009*
Countermovement Depth (in)	-8.902 ± 1.3	-12.38 ± 1.3	0.000*

***Statistically significant at $p < 0.05$*

Discussion

This study aimed to determine whether countermovement jump (CMJ) force-time curve shape (unimodal vs. bimodal) differentiates performance characteristics in Division I female basketball athletes. The results indicate distinct neuromechanical strategies associated with each curve profile, with several significant differences in kinetic variables and movement timing.

Bimodal jumpers exhibited longer braking and propulsive phases and longer total time to takeoff (TTT). These findings are consistent with prior work by Kennedy et al. (4), who reported similar phase duration differences and suggested that prolonged eccentric and concentric durations may allow greater engagement of stretch-shortening cycle (SSC) mechanisms. Notably, our study extends these findings by showing that bimodal jumpers also demonstrated greater relative braking force and braking power, variables not previously emphasized in Kennedy's work. These findings may reflect enhanced eccentric loading or force attenuation capabilities in this group.

Furthermore, performance trends in our study aligned with those reported by Peng et al. (2), who found higher jump height, greater reactive strength index-modified (RSI-mod), and deeper joint flexion angles in double peak curve athletes compared to single peak counterparts. Our data similarly showed significantly higher jump height, RSI-mod, and countermovement depth in bimodal jumpers. Peng et al. interpreted these outcomes as reflective of a more effective and coordinated CMJ strategy, facilitated by greater hip, knee, and ankle angular displacement

and improved proximal-to-distal extension sequencing. Although the present study did not assess kinematic joint displacement directly, the increased CMD in our bimodal athletes supports the interpretation that deeper movement strategies may aid force production.

However, unlike Peng et al., who observed no significant difference in time to take off between curve types, our data showed longer TTT in bimodal jumpers. This discrepancy could be attributed to differences in sample characteristics. Our cohort consisted of trained basketball athletes, whereas Peng's involved college students with no reported prior sports background. The athletic context may necessitate more deliberate countermovement strategies that increase temporal durations.

Unimodal athletes in our study showed higher average relative propulsive force, consistent with a stiffer, more rapid force application strategy. This aligns with Kennedy et al.'s (4) interpretation of unimodal curves as representative of faster SSC transitions and reduced joint excursion. Shallower CMD in unimodal athletes further supports the hypothesis that this group relies on brief eccentric loading to minimize movement duration while preserving output.

It is important to note that while bimodal athletes achieved higher peak outputs in several metrics, including power and jump height, this does not imply biomechanical superiority. Kennedy et al. (4) cautioned against interpreting bimodal patterns as more efficient, noting that variability in center of mass displacement and delayed force production may reflect compensatory or suboptimal strategies in some populations.

This study is not without limitations. The small sample size may limit generalizability, and curve classification was based on visual interpretation, which could introduce subjectivity. Additionally, this study was cross-sectional and focused exclusively on vertical CMJ

performance. Other research surrounding this subject has also proposed the idea that cueing may compound the results of countermovement jump performance and strategy. In Kennedy et al., athletes were cued to “Jump as high as possible,” while in Peng et al., individuals were instructed to “jump as fast and as high as possible.”

Future research should explore how training history, muscle-tendon properties, various cueing, and sport-specific demands influence curve expression and whether these profiles correspond to injury risk or training responsiveness.

Overall, this study's results highlight that the CMJ curve shape encapsulates meaningful differences in force-time strategies, with implications for profiling and individualized training. Both unimodal and bimodal patterns demonstrate viable neuromuscular approaches, and interpretation should be context-specific rather than prescriptive.

Acknowledgements

University of Pittsburgh Women’s Basketball

References

1. Kipp K, Kim H. Muscle-Specific Contributions to Vertical Ground Reaction Force Profiles During Countermovement Jumps: Case Studies in College Basketball Players. *Journal of Strength and Conditioning Research*. Published online February 7, 2023. doi:10.1519/JSC.0000000000004478

- 225 2. Peng HT, Song CY, Chen ZR, Wang IL, Gu CY, Wang LI. Differences Between Bimodal and
226 Unimodal Force-time Curves During Countermovement Jump. *Int J Sports Med*.
227 2019;40(10):663-669. doi:10.1055/a-0970-9104
- 228 3. McHugh MP, Hickok M, Cohen JA, Virgile A, Connolly DAJ. Is there a biomechanically
229 efficient vertical ground reaction force profile for countermovement jumps? *Transl Sports Med*.
230 2021;4(1):138-146. doi:10.1002/tsm2.200
- 231 4. Kennedy RA, Drake D. Is a Bimodal Force-Time Curve Related to Countermovement Jump
232 Performance? *Sports*. 2018;6(2):36. doi:10.3390/sports6020036

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